

ECE3337 Trigonometry Handout

$$\sin x = \pm \cos(x \mp 90^\circ)$$

$$\cos x = \pm \sin(x \pm 90^\circ)$$

$$\sin x = -\sin(x \pm 180^\circ)$$

$$\cos x = -\cos(x \pm 180^\circ)$$

$$\sin(-x) = -\sin x$$

$$\cos(-x) = \cos x$$

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$2 \sin x \sin y = \cos(x - y) - \cos(x + y)$$

$$2 \sin x \cos y = \sin(x + y) + \sin(x - y)$$

$$2 \cos x \cos y = \cos(x + y) + \cos(x - y)$$

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = 1 - 2 \sin^2 x$$

$$\sin x + \sin y = 2 \sin \left(\frac{x + y}{2} \right) \cos \left(\frac{x - y}{2} \right)$$

$$\sin x - \sin y = 2 \cos \left(\frac{x + y}{2} \right) \sin \left(\frac{x - y}{2} \right)$$

$$\cos x + \cos y = 2 \cos \left(\frac{x + y}{2} \right) \cos \left(\frac{x - y}{2} \right)$$

$$\cos x - \cos y = -2 \sin \left(\frac{x + y}{2} \right) \sin \left(\frac{x - y}{2} \right)$$

$$e^{jx} = \cos x + j \sin x \quad (\text{Euler's identity})$$

$$\sin x = \frac{e^{jx} - e^{-jx}}{2j}$$

$$\cos x = \frac{e^{jx} + e^{-jx}}{2}$$

Most of these formulas can be derived from Euler's identity, for example:

$$e^{j(A-B)} = \cos(A-B) + j \sin(A-B)$$

$$e^{jA} e^{-jB} = (\cos A + j \sin A) \times (\cos B - j \sin B)$$

$$= (\cos A \cos B + \sin A \sin B) + j(\sin A \cos B - \cos A \sin B)$$

Comparing the real and imaginary terms, we get:

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$